GOUTHAM GUNASEKARAN

Technical Lead | Automotive Embedded Software | Application SWC Developer

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Location: Coimbatore / Chennai, Tamil Nadu, India | Nationality: Indian

Open to relocation across Europe | Visa sponsorship required

PROFESSIONAL SUMMARY

Automotive Embedded Software Technical Lead with 10+ years of experience in AUTOSAR-based ECU application development, Body Electronics, HVAC climate control, SDV features, and industrial PLC/SCADA automation. Currently leading application SWC development for Mahindra BEV 6e and BE 9e electric vehicles, delivering front/rear lamp welcome/goodbye animations, SDV surprise message features, and dual-zone manual HVAC climate control module software. Expert in Embedded C, state machine design, CAN/CAN FD, BSW configuration, ASPICE-compliant SDLC, supplier coordination, and cross-functional team leadership.

LANGUAGES

English (Full Professional Proficiency) | Deutsch (Beginner / A1) | Tamil (Native)

CORE TECHNICAL SKILLS

Programming: Embedded C, C++, MicroPython, Python, MATLAB/Simulink

Domains: Body Electronics, HVAC / Climate Control, SDV Features, Lighting (LED Animations), Electric Power Steering, Industrial Automation

Architecture: AUTOSAR (Classic), State Machine Design, 3-Layer SWC Architecture, RTE Configuration & Generation

BSW Modules: NVM Stack (NvM, MemIf, Fee/Ea), COM Stack (Com, PduR, CanIf), Crypto Stack, MCAL, OS

Protocols: CAN, CAN FD, LIN, SPI, UART, I2C, ADC, PWM

AUTOSAR Tools: DaVinci Configurator Pro, DaVinci Developer, EB tresos, Enterprise Architect

Testing & Analysis: VectorCAST, Polyspace (MISRA C), CANoe, VFlash, Logical Analyzer, Oscilloscope

PLC / Automation: Siemens, Allen-Bradley, ABB, Delta PLCs, SCADA, HMI Development

Requirements: IBM DOORS, Microsoft Visure, Reqtify (Traceability)

PCB / Schematic: Altium Designer, EasyEDA

IDEs & Build: IAR Embedded Workbench, MentorGraphics VSB

Project Tools: JIRA, Confluence, Git

Scripting / Tooling: Python Tkinter (Windows automation tools)

Standards: ASPICE (SWE.1–SWE.6), MISRA C, Automotive SDLC

PROFESSIONAL EXPERIENCE

Mahindra and Mahindra Ltd | Chennai / Coimbatore, India

Lead Engineer – Application Software | Mahindra Software Defined Vehicle Centre (MSDVC) | MAY 2023 – Present

· Lead application software development and delivery for Mahindra BEV 6e and BE 9e electric vehicle platforms at MSDVC, coordinating with cross-functional teams including system, hardware, and validation engineers.

· Developed and delivered front and rear lamp welcome/goodbye animation features using scalable state machine architecture in Embedded C, enabling dynamic LED lighting sequences across Mahindra electric vehicle product lines.

· Implemented SDV-based customizable surprise message animation feature, allowing over-the-air personalization of vehicle lighting behavior as part of Mahindra's Software Defined Vehicle strategy.

· Currently leading software development for the ETC Dual-Zone Manual HVAC Climate Control Module, designing application SWCs for temperature regulation, blower control, air distribution, and mode management across driver and passenger zones.

· Defined and standardized scalable state machine architecture methodologies for automotive application software, establishing reusable design patterns adopted across multiple ECU projects within the MSDVC division.

· Guided internal teams and external suppliers on common software development approaches, conducting technical reviews of architecture design, detailed design, and code to ensure ASPICE compliance.

· Managed end-to-end SDLC activities following ASPICE standards (SWE.1–SWE.6) including requirements analysis, architecture design, detailed design, coding, unit testing, and integration testing.

IQuants Engineering Solutions Pvt Ltd | Coimbatore, India

Senior Embedded Engineer | FEB 2020 – APR 2023

· Developed and maintained AUTOSAR-compliant embedded software for automotive ECU projects, handling end-to-end responsibility from requirements analysis through integration testing and production delivery.

· Engineered bare-metal firmware for microcontroller-based embedded systems, optimizing memory footprint and execution timing for resource-constrained automotive targets.

· Configured AUTOSAR BSW modules including NVM Stack (NvM, MemIf, Fee), COM Stack (Com, PduR, CanIf) using DaVinci Configurator Pro and EB tresos for multiple ECU variants.

· Integrated the AUTOSAR EEPROM NVM stack into the flash bootloader, enabling non-volatile data persistence and reliable ECU reflashing across bootloader stages.

· Performed RTE generation and configuration using DaVinci Developer, ensuring seamless inter-SWC communication and correct port/interface mappings between application and BSW layers.

· Designed hardware schematics and corresponding embedded software for prototype ECU boards, supporting rapid prototyping and hardware-software co-development.

· Debugged complex embedded software defects on target hardware using CANoe, logical analyzers, and oscilloscopes, performing root cause analysis for timing-critical and communication protocol issues.

Self-Employed (Freelance) | Coimbatore, India

Solar Panel Installation Engineer | FEB 2017 – JAN 2020

· Operated as an independent freelance engineer specializing in residential and commercial solar panel installation, managing end-to-end project execution from site assessment through commissioning.

· Performed electrical system design and wiring for solar PV installations, applying knowledge of power electronics, circuit protection, and electrical safety standards.

· Managed client relationships, project timelines, material procurement, and on-site installation crews, developing strong project management and stakeholder coordination skills.

STEPS Knowledge Services Pvt Ltd | Coimbatore, India

Assistant Engineer | MAY 2015 – JAN 2017

· Designed and developed PLC-based industrial automation solutions using Siemens, Delta, Allen-Bradley, and ABB programmable logic controllers for manufacturing plant control systems.

· Delivered end-to-end plant automation for VV Nationals Industry, engineering the complete control system for Hopper, wet chew vibrator, and iron ore extraction line from PLC programming through commissioning.

· Developed SCADA systems and HMI interfaces for real-time plant monitoring, alarm management, and operator control, enabling centralized visibility across multiple production stages.

· Designed, coded, and maintained AUTOSAR application SWCs following the 3-layer architecture (Application, RTE, BSW) for the Audio Amplifier ECU project, including RTE configuration and integration for application APIs.

· Developed and executed integration test cases derived from architecture design documents, achieving 100% requirements-to-test traceability for the Audio Amplifier ECU delivery.

ACME Systems | Coimbatore, India

Embedded Developer | JUL 2012 – MAR 2015

· Developed application-layer software components (SWCs) in Embedded C for automotive ECU projects, implementing control algorithms and diagnostic routines per customer specifications.

· Contributed to the Electric Power Steering (EPS) project for Nexteer Automotive, developing auto-generated and hand-coded MATLAB/Simulink models for steering control algorithms.

· Conducted static code analysis using Polyspace, ensuring full compliance with MISRA C coding guidelines and achieving zero critical violations in delivered production code.

· Designed hardware schematics and PCB layouts using Altium Designer for embedded controller prototypes, supporting rapid hardware iteration and design validation.

KEY PROJECTS

Front & Rear Lamp Welcome/Goodbye Animations – Mahindra BEV 6e / BE 9e

Technologies: AUTOSAR, Embedded C, State Machine Architecture, CAN FD

· Designed and implemented scrolling text animation and dynamic LED lighting algorithms for front and rear lamps on Mahindra's flagship electric vehicles.

· Delivered SDV-based surprise message feature enabling OTA-customizable animation sequences for personalized vehicle lighting.

Dual-Zone Manual HVAC Climate Control Module

Technologies: AUTOSAR, Embedded C, CAN/CAN FD, State Machine Design

· Leading application software development for ETC dual-zone manual climate control, managing temperature, blower, and air distribution logic for driver/passenger zones.

VV Nationals Industry – Complete Plant Automation

Technologies: Siemens PLC, SCADA, HMI, Industrial Automation

· Delivered end-to-end automation for iron ore extraction plant including Hopper and wet chew vibrator systems, from PLC programming through commissioning and operator training.

Electric Power Steering (EPS) – Nexteer Automotive

Technologies: AUTOSAR, MATLAB/Simulink, Polyspace, MISRA C

· Developed auto-code and hand-code Simulink models for EPS control algorithms with MISRA C compliance via Polyspace static analysis.

Audio Amplifier ECU Development

Technologies: AUTOSAR RTE, ASPICE, Embedded C

· Configured AUTOSAR RTE integration, developed integration test cases from architecture design, and ensured 100% requirements traceability.

EDUCATION

Bachelor of Engineering (B.E.) – Electronics & Communication Engineering

VRS College of Engineering and Technology – Anna University

CERTIFICATIONS

- National Instruments LabVIEW Core-1
- Vector Tools Training Program (CANoe, CAPL, UDS)